Phase 2 — Requirement Analysis Phase

Student Enrollment & Course Management System on Salesforce

Academic Year 2026-2027

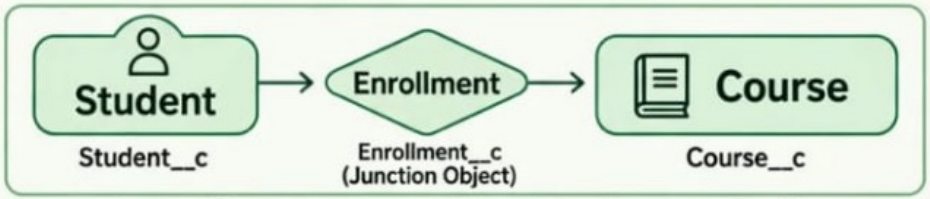
CUSTOMER JOURNEY

Student Applies → Admin Verifies → Student Enrolls in Course → Faculty Updates Grades → Student Views Report

FUNCTIONAL REQUIREMENTS

- Student Registration
- Course Creation
- Course Enrollment
- Fee Payment
- Attendance
- Grade Management

DATA FLOW DIAGRAM



Student and Course are connected via the junction object Enrollment to manage the many-to-many relationship.

TECHNOLOGY STACK

- Salesforce Custom Objects
- Flows for Automation
- Apex Triggers for Complex Logic
- LWC for UI

salesforce

Academic Year 2026–2027

Student Enrollment and Course Management System

PHASE 2 — REQUIREMENT ANALYSIS PHASE | DETAILED EXPLANATION

DETAILED CUSTOMER JOURNEY

5 Steps from Application to Reporting

1. **Student Applies**
   Form submission via online portal
2. **Admin Verifies Documents**
   Review submitted documents, validate eligibility
3. **Student Enrolls in Course**
   Seat allocation based on capacity & prerequisites
4. **Faculty Updates Grades & Attendance**
   Input grades and log attendance records
5. **Student Views Report Dashboard**
   Access performance, grades, and enrollment dashboard

FUNCTIONAL REQUIREMENTS — DEEP DIVE

Student Registration

- **Auto ID Generation:** System auto-generates unique **Student ID** (STU-YY YYYY-###)

Course Creation

- **Credits:** Define credit hours for the course
- **Capacity:** Set maximum seat capacity
- **Preqriuisites:** Specify required preqreqisite courses

Course Enrollment

- **Junction Logic:** Many-to-Many relationship between Students and Courses
- Validates seat availability & preqreqisite checks before enrollment

Fee Payment

- **Tracking Status:** Pending | Partial | Paid | Refunded
- Invoice generation & payment history tracked

Key Focus: Data Accuracy | Complience | Transparency

Designed for Academic Admins • Scalable • Secure • Salesforce-Powered

Student Enrollment & Course Management System

Academic Year 2026–2027 • Detailed Technical Analysis & Design

NON-FUNCTIONAL REQUIREMENTS

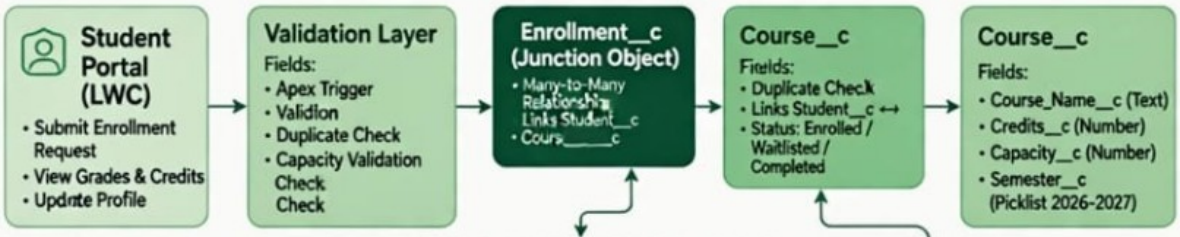
Security

- Role-Based Access Control (RBAC) via Profiles & Permission Sets
- Field-Level Security for sensitive data: GPA, Email, Phone
- Oauth 2.0 + Multi-Factor Authentication enforced
- Audit Trail enabled on all Custom Objects

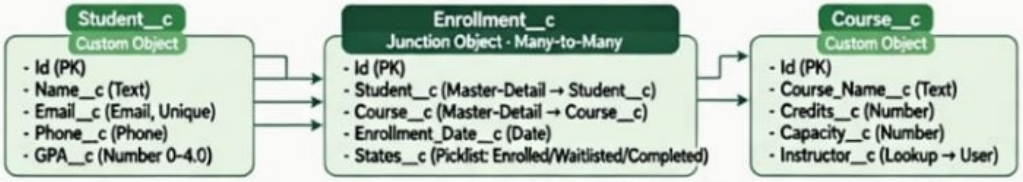
Performance

- Optimiled SOQL with selective filters & indexing
- Bulk processing support for 2000+ enrollment records
- Platform Cache utilized for course catalog data
- Supports 50K+ students & 1000+ courses stalability
- Asynctroness processing via Queueable Apex
- Partitioned data strategy for historical enrollment records

DETAILED DATA FLOW DIAGRAM



ER DIAGRAM VISUALIZATION



Enrollment__c serves as the junction connecting Student__c and Course__c in a many-to-many relationship

TECHNOLOGY STACK DEEP DIVE

Custom Objects, Fields & Relationships

- **Student__c**
  Email (Email, Unique),
  Phone, GPA (0–4.0),
- **Course__c**
  Credits__c,
  Capacity__c, Semester__c (Picklist 2026-2027)
- **Enrollment__c**
  Junction object with Master-Detail relationships

Relationships

- **Student__c → Enrollment__c:** Master-Detail (Rollup: Total_Enrollments__c)
- **Course__c → Enrollment__c:** Master-Detail (Rollup: Seats ll_c, Remaining_Capacity__c)
- **Course__c → Instructor:** Lookup → User (Instructor)

Flows & Automation

- **Screen Flow:** Student Self-Enrollment Flow (interactive UI)
  Autolaunched Flow:
  Daily Waitlist Notification & Capacity Check
- **Record-Triggered Flow:** Auto-update GPA on Grade Change

Apex Triggers

- **CapacityValidationTrigger:** Prevent enrollment when Course__c Capacity__c is reached
- **DuplicateCheckTrigger:** Prevent duplicate Student+Course enrollment
- **Before Insert/Update:** Validate GPA within 0.0–4.0 range

LWC STUDENT PORTAL COMPONENTS

student-portal-dashboard

- Display enrolled courses & current GPA
- Show credit summary & academic status
- Pending actions & notifications

course-catalog-browser

- Search & filter by course, credits, semester
- View course details, capacity & instructor
- Enroll bution with capacity check

enrollment-form

- Self-enrollment form with validation
- Confirmation screen & enrollment status
- Waitlist option if course is full

KEY BENEFITS

Data Integrity
Duplicate prevention & validation ensure accurate, reliable enrollment data

Optimized Performance
Scalable architecture supports thousands of records with efficient query patterns

Enhanced Student Experience
Self-service LWC portal enables seamless enrollment & grade visibility for students

Security & Compliance
Field-level security, audit logging, and role-based access meet academic compliance